

2nd yr. 3rd sem

Student Roll No. _____

ALIAH UNIVERSITY

B. Tech. Semester: III, Civil Engineering

Odd Semester Examination Dec-2022

Subject Code & Name: (CENUGPC03), Building Materials

Time: 3 Hours

Total Marks: 80

Instruction: (1) Figures to the right indicate full marks.
(2) Assume suitable data if required.
(3) Question-1 is compulsory.

Que: -1 Answer the following questions

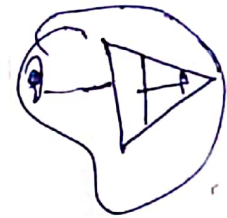
2×10=20

- ☒ A Write a note on English Bond with neat sketch.
- ☒ B Define terms-Headroom, Newel posts, Handrail, landing.
- ☐ C The bulking of sand is 10%, if the mix ratio is 1:2:4. What would be the actual volume of sand used per unit of cement.
- ☒ D What is the different between paints and varnishes?
- ☒ E Describe the tests performed to check the quality of bricks.
- ☒ F Discuss the following defects- Shakes and Knots.
- ☒ G What is the composition of cement clinker.
- ☐ H Describe the different varieties of lime
- ☒ I Classify the concrete on the basis of Bulk density and grade of cement concrete.
- ☒ J What proportions of mortar will you recommend in the following cases. (a) lime concrete in foundation (b) brick work in foundation

Que: -2 Attempt any Six.

10×6=60

- ☒ A What is mortar? Briefly describe the various types of mortars.
- ☒ B Write a detail note on Classification of bricks.
- ☐ C Briefly describe the Specific gravity and water absorption test on aggregate.
- ☐ D Provide a brief description of wood preservation practices.
- ☐ E Enlist and explain the different types of doors used in public and residential buildings with neat sketches.
- ☐ F Briefly describe the Split tensile test and Compacting factor test for concrete.
- ☒ G Discuss the different types of cement in detail.



Odd Semester Examination 2

Subject Code: CENUGPC

Subject Name: Surveying and Geomatics

B. Tech Civil Engineering III Semester

Full Marks: 80 (8x10)

Attempt Any 8 Questions

1. Explain various terms with a suitable diagram:

- a) Back Sight b) Foresight c) Line of Coll

2. What are the different types of survey based on the instrument and nature of survey?

3. What are the different errors in levelling? Calculate the reduced height of the instrument using the following dataset:

A=Instrument station; D= horizontal distance between A to the 500mm

$\angle BAC = 15^\circ$; Bench Mark=100m; Staff Reading=1m; Fore

DS.

4. Explain the physics of Remote Sensing using the elec

5. What do you mean by triangulation? Describe well and

6. State the working principles of the Global Navigation

Global Positioning System (GPS) and explain them with

7. Based on the concept of properties of digital images, t

- a) Spatial Resolution b) Spectral Resolution

8. What are the advantages and disadvantages of raster
play a crucial role in flood monitoring?

9. How does Normalized Difference Vegetation Inde
vegetation health and crop health monitoring for any re

10. What do you mean by pure and mixed pixels? Explain

1. Answer any four questions: (20X4=80)

- a) What are the major streams of the Civil Engineering discipline? Briefly explain the major stream as per their application in the society. (5+15=20)
- b) Explain the importance of surveying in Civil Engineering. Elaborate the different surveying techniques applied in Civil Engineering. (5+15=20)
- c) Mention the different types of concrete and their application for various civil works. What is meant by characteristic compressive strength of concrete. What are the different properties of concrete? (10+5+5=20)
- d) What are the considerations for selecting the tread and rise of a staircase? Differentiate between a dog legged staircase and a quarter turn staircase. What is meant by fineness, crushing strength, soundness and setting time of cement? Differentiate between mild steel and high carbon steel. What is meant by curing of concrete? Differentiate between artificial and natural seasoning of timber. (3+3+6+3+3+2=20)
- e) What is the difference between accident and disaster? What are the basic steps to be taken during a disaster? What are the environmental impacts of a disaster? Enumerate the basic flood control measures. How can a forest fire be prevented? (4+4+4+4+4=20)
- f) Explain the utmost importance of IS Codes in the discipline of Civil Engineering with reference to three relevant codes pertinent to three streams of Civil Engineering. What are the basic responsibilities of a town planner? What is ocean engineering? What is meant by repairing and retrofitting of a structure? (5+5+5+5=20)

Mild steel specific gravity is 2.8, It is easy to welded, It is strong in compression and tension both, It is use in RCC works.

Carbon steel It has 0.5 to 1.5% Carbon, It is very hard to welded, it is stronger in compression than tension, It is use in prestress concrete works.

Odd Semester Examination, 2022
UG Section
Sub: Indian Constitution (UCCUGAU03)

Full Marks: 80

Time: 3hrs

1. Answer the following questions: (within 30 words each)

2x5=10

- a. Define 'Fundamental Rights'.
- b. What Principles were added to the Constitution of India by the 42nd Amendment Act?
- c. Discuss Article 21A of the Constitution of India.
- d. Discuss any two functions of the Speaker of the Lok Sabha.
- e. Discuss the composition of the Upper House of the Indian Parliament.

2. Write any six questions: (within 100 words each)

5x6=30

- a. Write a short note on the writs of Habeas Corpus and Mandamus.
- b. Discuss in brief Articles 23 and 24 as enumerated in Part III of the Constitution of India.
- c. Enumerate the differences between the Fundamental Rights and the Directive Principles of State Policy.
- d. Describe the election procedure of the President of India.
- e. Mention Five Fundamental Duties mentioned in our Constitution.
- f. Discuss the method of appointment of the Chief Minister of Indian States.
- g. Discuss the position of the Governor of an Indian state.
- h. Critically discuss the Right to Freedom of Religion.

3. Write any four questions: (Within 350 words each)

10x4= 40

- a. Discuss Article 19 as enumerated in the Constitution of India and point out its reasonable restrictions.
- b. "The Constitution of India is neither purely federal nor unitary but a combination of both". Do you agree with this view? Give reasons in support of your answer.
- c. Discuss the powers and position of the President of India.
- d. Analyse the relation between the two houses of the Parliament.
- e. Discuss the composition and powers of the Supreme Court of India.

ALIAH UNIVERSITY

Department of Civil Engineering
End-Semester Examination- December, 2022

Semester: 3rd

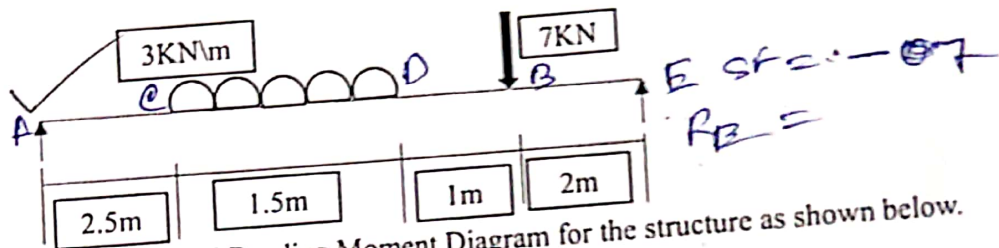
Subject Code: CENUGPCO1

Time: 3 hrs.

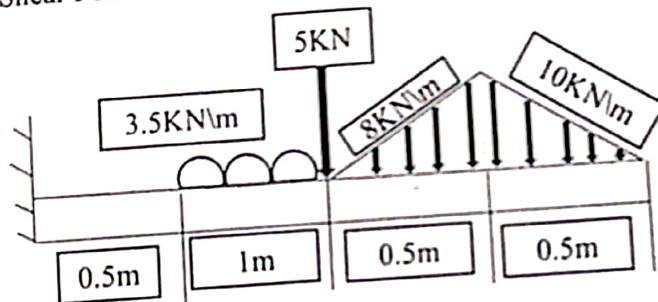
Year: 2nd
Subject: Mechanics of Materials
Full Marks: 80

Attempt all question and assume suitable data if required.
All Questions are of Equal Marks

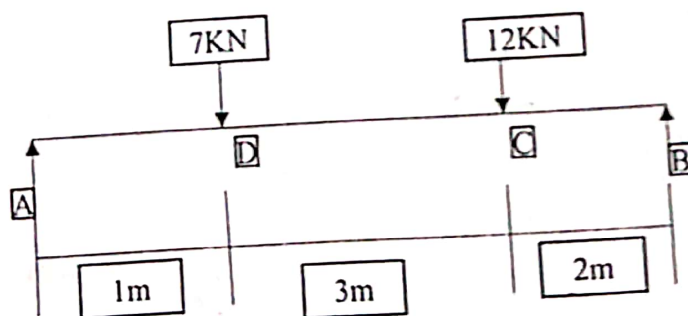
1. Draw Shear Force and Bending Moment Diagram for the structure as shown below.



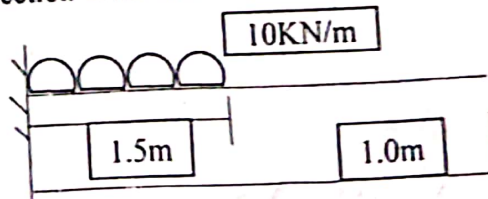
2. Draw Shear Force and Bending Moment Diagram for the structure as shown below.



3. Calculate the deflections of the beam under the loads C and D. Take, $E = 200 \text{ GPa}$ and $I = 160 \times 10^6 \text{ mm}^4$.



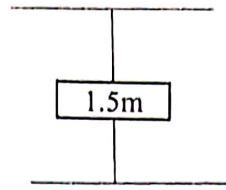
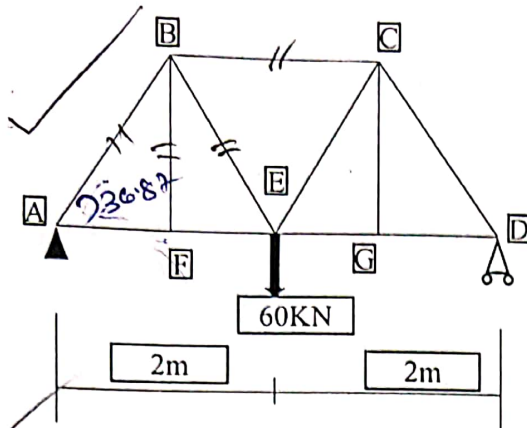
4. Calculate the slope and deflection at the free end of the cantilever. Take, $EI = 1.9 \times 10^{12} \text{ N-mm}^2$.



5. Analysis the Truss as drawn below

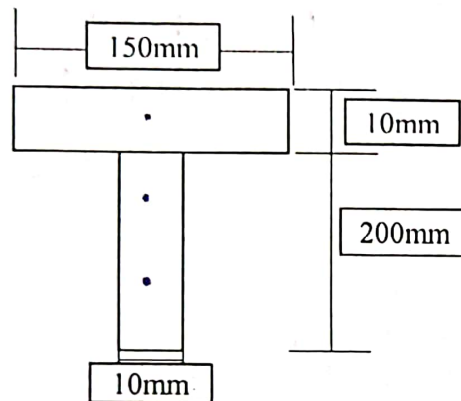
∴ Therefore,

= .



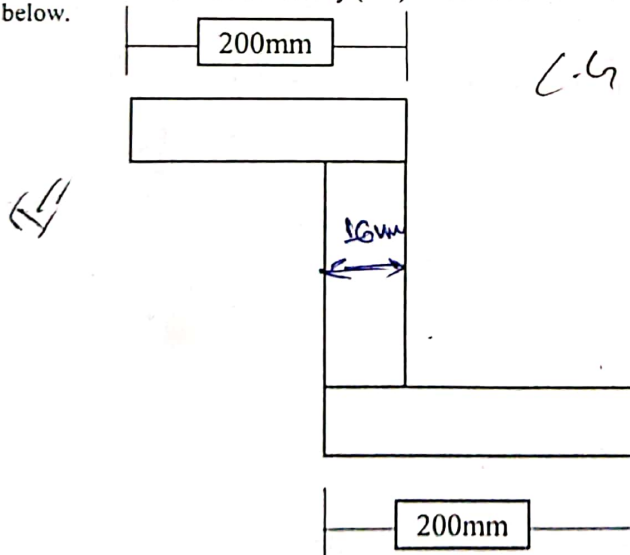
$$\tan^{-1} = \frac{1.5}{2} = 36.87^\circ$$

6. Calculate the Centre of Gravity (CG) and Moment of Inertia (MI) about the CG for the section as shown below.



$$\begin{aligned} a_1 &= 3200 \\ a_2 &= 4800 \\ a_3 &= 3200 \\ y_1 &= 8 \\ y_2 &= 166 \\ y_3 &= 324 \end{aligned}$$

7. Calculate the Centre of Gravity (CG) and Moment of Inertia (MI) about the CG for the section as shown below.



$$C.G. = \bar{y} = 166$$

$$\begin{aligned} (I_{xx})_1 &= 7.4 \times 10^7 \\ (I_{xx})_2 &= 3.7 \times 10^7 \\ (I_{xx})_3 &= 7.9 \times 10^7 \\ I_{xx} &= I_{CG} = I_{xx1} + I_{xx2} + I_{xx3} \\ &= 7.9 \times 10^7 \end{aligned}$$

18. A mild steel column of 50 mm diameter is hinged at both at the ends. Find the crippling load for the column, if its length is 3m. Take E for the column material as 200 Gpa.

$$21.418 \text{ kN}$$

$$(1) = -4 < 0$$

$$(2) = 370$$

$$x^2 + 2x - 2$$

$$= 2x + 2$$

$$3x^2$$

$$f(1) = 170$$

$$f(0) = -2 < 0$$

ALIAH UNIVERSITY
THIRD SEMESTER EXAMINATION, 2022
PAPER MATUGBS03 (ENGINEERING MATHEMATICS-III)(MEN-CEN)
Full Marks 80 Time 150 Minutes
Unless Otherwise Stated Notations Carry their Usual Meanings
Use separate Answer Scripts for Each Group.

GROUP - A

Answer any FOUR questions.

10 × 4

1. (a) Find a real root of the equation $x^3 - x + 1$ by the method of bisection correct three significant digits. (7)
- (b) Calculate the positive root of the equation $x^2 + 2x - 2 = 0$ correct up to significant figures by the Newton-Raphson method. (3)
2. (a) Use Newton-Raphson method to find real root of the equation $x^3 - 5 = 0$ correct three significant digits and hence find the value of $\sqrt[3]{5}$. (6)
- (b) Given the following set of values of $y = f(x)$ (4)

x	2	4	6	8	10
y	5	10	17	29	50

form the diagonal difference table and find the value of $\Delta f(6)$, $\Delta^2 f(6)$.

3. (a) Calculate $f(3.5)$ using the table by using Lagrangian formula (6)

x	0.3	0.5	0.6
f(x)	0.6179	0.6195	0.7257

- (b) Form the interpolation polynomial for the function $y = f(x)$ given by the table (4)

x	-1	1	4	6
y	1	-3	21	127

4. (a) Evaluate $y(1.1)$ using Runge-Kutta method of order four for the initial value problem (5)

$$\frac{dy}{dx} = x^2 + y^2, \quad y(1) = 0.$$

- (b) Apply Simpson's three-eighth rule, find out the value of $\int_{1.00}^{1.30} \sqrt{x} \, dx$. (5)

5. (a) Using trapezoidal rule evaluate (approximately) the integral

$$\int_0^1 (4x - 3x^2) dx \quad (7)$$

by taking $n = 10$.

- (b) Find out the solution by using Weddle's rule

x	0	1	2	3	4	5	6
$f(x)$	1	0.5	0.2	0.1	0.585	0.0385	0.027

6. (a) Solve the equation by Picard's method: (3)

$$\frac{dy}{dx} = xy + 1 \quad \text{with } y(0) = 1. \quad (5)$$

- (b) Using Euler's method, solve the differential equation

$$\frac{dy}{dx} = xy, \quad \text{with } y = 1 \quad \text{when } x = 0 \quad (5)$$

to find $y(1.0)$ by taking $h = 0.2$.

7. Evaluate

$$\int_0^1 \frac{dx}{1+x^2}$$

using Simpson's one-third rule and hence find out the value of π . (7+3)

GROUP - B

1. Answer any **one** question from the following Questions:

- (a) Form the partial differential equation by eliminating the arbitrary constant from

$$\log_e(az - 1) = x + ay + b. \quad (5)$$

- (b) Form a partial differential equation by eliminating the arbitrary function ϕ from

$$lx + my + nz = \phi(x^2 + y^2 + z^2). \quad (5)$$

2. Answer any **one** question from the following Questions:

- (a) Solve by Lagrange's method: here $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$. (5)

$$\frac{y^2 z}{x} p + xzq = y^2.$$

(b) Solve by Lagrange's method:

$$y^2p - xyq = x(z - 2y).$$

[5]

3. Answer any **one** question from the following questions:

(a) Solve by Charpit's method:

[5]

$$px + qy = pq.$$

(b) Find the complete integral of the following partial differential equation by Charpit's method:

[5]

$$2xz - px^2 - 2qxy + pq = 0.$$

4. Answer any **one** question from the following Questions:

(a) Deduce the one-dimensional heat equation.

[5]

(b) Using the method of separation of variables, solve the following initial value problem:

[5]

$$\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u \quad \text{if } u(x, 0) = 6e^{-3x}.$$

5. Answer any **one** question from the following Questions:

(a) Find the Fourier series of the function:

[10]

$$f(x) = x - x^2, \quad -\pi < x < \pi.$$

Hence deduce that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}.$

(b) Obtain the Fourier expansion of the function:

[10]

$$f(x) = x \sin x \quad \text{in } -\pi \leq x \leq \pi.$$

Hence deduce that $\frac{\pi}{4} = \frac{1}{2} + \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots.$

6. Answer any **one** question from the following Questions:

(a) Find the Fourier Transformation of the function $F(x)$ given by

[5]

$$F(x) = \begin{cases} x^2, & |x| \leq a \\ 0, & |x| > a \end{cases}$$

(b) Find $F(x)$, if its Fourier Sine transform is $\frac{s}{1+s^2}.$

[5]

7. Answer any **one** question from the following Questions:

(a) Find the extremals of the functional

[5]

$$\int_0^\pi \left[4y \cos x + \left(\frac{dy}{dx} \right)^2 - y^2 \right] dx$$

that satisfies the boundary conditions $y(0) = y(\pi) = 0.$

(b) Find the curve connecting the given points A and B , which is traversed by a particle sliding from A to B in the shortest time (friction and resistance of the medium are ignored).

[5]